

EDUCATION	University of Waterloo	Waterloo, Canada
	<i>Ph.D. in Computer Science Advisor: Prof. Jimmy Lin</i>	Sept. 2020 – Present
	Hong Kong University of Science and Technology	Hong Kong SAR China
	<i>B.E in Computer Science Thesis Advisor: Prof. Lei Chen</i>	Sept. 2015 - Jan. 2020
	University of California, Los Angeles	Los Angeles, United States
	<i>International Exchange Program</i>	Jun. 2017 – Aug. 2017
EMPLOYMENT	Google DeepMind New York City, United States	Nov. 2023 – Aug. 2024
	<i>Student Researcher Host: Dr. Jing Lu, Manager: Dr. Donald Metzler</i>	
	Cohere Toronto, Canada	May 2023 – Oct. 2023
	<i>Research Intern Host: Dr. Sebastian Hofstätter, Dr. Patrick Lewis</i>	
	Max Planck Institute for Informatics Saarbrücken, Germany	Feb. 2020 – Aug. 2020
	<i>Visiting Scholar Fellowship Host: Prof. Andrew Yates</i>	
	NAVER Clova AI Lab Seongnam, South Korea	Sept. 2018 – Dec. 2018
	<i>AI Research Intern Supervisor: Prof. Lei Chen, Prof. Sung Kim</i>	
PUBLICATIONS	UROP Hong Kong SAR, China	Sept. 2017 – Aug. 2018
	<i>Undergraduate Student Researcher Supervisor: Prof. Samuel Chow, Prof. Gary Shueng Han Chan</i>	
	[1] The Impact of Incidental Multilingual Text on the Cross-Lingual Transferring in Monolingual Retrieval	
	Andrew Liu*, Edward Xu*, Xinyu Zhang , Jimmy Lin	
	<i>In the 47rd European Conference on Information Retrieval</i>	Apr. 2025
	[2] Rank-without-GPT: Building GPT-Independent Listwise Rerankers on Open-Source Large Language Models	
	Xinyu Zhang , Sebastian Hofstätter, Patrick Lewis, Raphael Tang, Jimmy Lin	
	<i>In the 47rd European Conference on Information Retrieval</i>	Apr. 2025
	[3] Words Worth a Thousand Pictures: Measuring and Understanding Perceptual Variability in Text-to-Image Generation	
	Raphael Tang, Xinyu Zhang , Lixinyu Xu, Yao Lu, Wenyan Li, Pontus Stenetorp, Jimmy Lin, Ferhan Ture	
	<i>In the 2024 Conference on Empirical Methods in Natural Language Processing</i>	
	[Outstanding Paper Award]	
	[4] FoodieQA: A Multimodal Dataset for Fine-Grained Understanding of Chinese Food Culture	
	Wenyan Li, Xinyu Zhang , Jiaang Li, Qiwei Peng, Raphael Tang, Li Zhou, Weijia Zhang, Guimin Hu, Yifei Yuan, Anders Søgaard, Daniel Hershcovich, Desmond Elliott	
	<i>In the 2024 Conference on Empirical Methods in Natural Language Processing</i>	
	[5] NoMIRACL: Knowing When You Don't Know for Robust Multilingual Retrieval-Augmented Generation	
	Nandan Thakur, Luiz Bonifacio, Xinyu Zhang , Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Boxing Chen, Mehdi Rezagholizadeh, Jimmy Lin	
	<i>In the 2024 Conference on Empirical Methods in Natural Language Processing</i>	
	[6] CELI: Simple yet Effective Approach to Enhance Out-of-Domain Generalization of Cross-Encoders.	
	Xinyu Zhang* , Minghan Li*, Jimmy Lin	
	<i>In the 2024 Conference of the North American Chapter of the Association for Computational Linguistics</i>	Jun. 2024

- [7] Found in the Middle: Permutation Self-Consistency Improves Listwise Ranking in Large Language Models
Raphael Tang*, **Xinyu Zhang***, Xueguang Ma, Jimmy Lin and Ferhan Ture
In the 2024 Conference of the North American Chapter of the Association for Computational Linguistics Jun. 2024
- [8] CIRAL: A Test Collection for CLIR Evaluations in African Languages.
Mofetoluwa Adeyemi, Akintunde Oladipo, **Xinyu Zhang**, Jimmy Lin, David Alfonso-Hermelo, Mehdi Rezagholizadeh, Boxing Chen, ... (17 authors)
In the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval Jul. 2024 **[Best Paper Nominee]**
- [9] MIRACL: A Multilingual Retrieval Dataset Covering 18 Diverse Languages
Xinyu Zhang*, Nandan Thakur*, Odunayo Ogundepo, Ehsan Kamalloo, David Alfonso-Hermelo, Xiaoguang Li, Qun Liu, Mehdi Rezagholizadeh, and Jimmy Lin
Transactions of the Association for Computational Linguistics, Sept. 2023
- [10] Toward Best Practices for Training Multilingual Dense Retrieval Models
Xinyu Zhang, Kelechi Ogueji, Xueguang Ma, Jimmy Lin
ACM Transactions on Information Systems 42 (2), 1–33, Sept. 2023
- [11] Spacerini: Plug-and-play search engines with Pyserini and Hugging Face
Christopher Akiki, Odunayo Ogundepo, Aleksandra Piktus, **Xinyu Zhang**, Akintunde Oladipo, Jimmy Lin, Martin Potthast
In the 2023 Conference on Empirical Methods in Natural Language Processing
- [12] GAIA Search: Hugging Face and Pyserini Interoperability for NLP Training Data Exploration
Aleksandra Piktus, Odunayo Ogundepo, Christopher Akiki, Akintunde Oladipo, **Xinyu Zhang**, Hailey Schoelkopf, Stella Biderman, Martin Potthast, Jimmy Lin
In the 2023 Conference of the Association for Computational Linguistics
- [13] Evaluating Embedding APIs for Information Retrieval
Ehsan Kamalloo, **Xinyu Zhang**, Odunayo Ogundepo, Nandan Thakur, David Alfonso-Hermelo, Mehdi Rezagholizadeh, Jimmy Lin
In the 2023 Conference of the Association for Computational Linguistics
- [14] Certified Error Control of Candidate Set Pruning for Two-Stage Relevance Ranking
Minghan Li*, **Xinyu Zhang***, Ji Xin, Hongyang Zhang, and Jimmy Lin
In the 2022 Conference on Empirical Methods in Natural Language Processing Dec. 2022 **[Oral]**
- [15] AfriCLIRMatrix: Enabling Cross-Lingual Information Retrieval for African Languages
Odunayo Ogundepo, **Xinyu Zhang**, Shuo Sun, Kevin Duh, and Jimmy Lin
In the 2022 Conference on Empirical Methods in Natural Language Processing, Dec. 2022
- [16] Squeezing Water from a Stone: A Bag of Tricks for Further Improving Cross-encoder Effectiveness for Reranking
Ronak Pradeep*, Yuqi Liu*, **Xinyu Zhang**, Yilin Li, Andrew Yates, and Jimmy Lin
In the 44rd European Conference on Information Retrieval, Apr. 2022
- [17] Mr. TyDi: A Multi-lingual Benchmark for Dense Retrieval
Xinyu Zhang, Xueguang Ma, Peng Shi, Jimmy Lin
In the 2021 Conference on Empirical Methods in Natural Language Processing Workshop on Multilingual Representation Learning, Nov. 2021 **[Oral]**
- [18] Comparing Score Aggregation Approaches for Pretrained Neural Language Models
Xinyu Zhang, Andrew Yates, and Jimmy Lin
European Conference on Information Retrieval (ECIR), 2021
- [19] Bag-of-Words Baselines for Semantic Code Search
Xinyu Zhang, Ji Xin, Andrew Yates, Jimmy Lin
Proceedings of the 1st Workshop on Natural Language Processing for Programming, 2021
- [20] A Little Bit Is Worse Than None: Ranking with Limited Training Data
Xinyu Zhang, Andrew Yates, and Jimmy Lin
In the 2020 Conference on Empirical Methods in Natural Language Processing Workshop on Simple and Efficient Natural Language Processing, Nov. 2020

- [21] Flexible IR Pipelines with Capreolus
Andrew Yates, Kevin Martin Jose, **Xinyu Zhang**, Jimmy Lin
In the 29th ACM International Conference on Information and Knowledge Management, Oct. 2020

PREPRINT

- [20] Tomato, Tomahto, Tomate: Measuring the Role of Shared Semantics among Subwords in Multilingual Language Models
Xinyu Zhang, Jing Lu, Vinh Q. Tran, Tal Schuster, Donald Metzler, Jimmy Lin
ArXiv 2411.04530
- [21] What do Llamas really think? Revealing preference biases in language model representations
Raphael Tang, **Xinyu Zhang**, Jimmy Lin, Ferhan Ture
ArXiv 2311.18812
- [22] Zero-shot listwise document reranking with a large language model
Xueguang Ma, **Xinyu Zhang**, Ronak Pradeep, Jimmy Lin
ArXiv 2305.02156
- [23] Simple yet effective neural ranking and reranking baselines for cross-lingual information retrieval
Jimmy Lin, David Alfonso-Hermelo, Vitor Jeronymo, Ehsan Kamalloo, Carlos Lassance, Rodrigo Nogueira, Odunayo Ogundepo, Mehdi Rezagholizadeh, Nandan Thakur, Jheng-Hong Yang, **Xinyu Zhang** (in alphabetical order) *ArXiv 2304.01019*
- [24] HAGRID: A human-LLM collaborative dataset for generative information-seeking with attribution
Ehsan Kamalloo, Aref Jafari, **Xinyu Zhang**, Nandan Thakur, Jimmy Lin
ArXiv 2307.16883
- [25] Better Than Whitespace: Information Retrieval for Languages without Custom Tokenizers
Odunayo Ogundepo, **Xinyu Zhang**, Jimmy Lin
ArXiv 2210.05481

MENTORING

Yuqi Liu. Undergrad and Master Student of UWaterloo. Published [14]	2021
Odunayo Ogundepo. Master Student of Uwaterloo. Published [13,25]	2022
Mofetoluwa Adeyemi. Master Student of UWaterloo. Published [3]	2023
Andrew Liu. Undergrad Student of UWaterloo. Paper Under Review	2024
Edward Xu. Undergrad Student of UWaterloo. Paper Under Review	2024

TALKS

SIGIR 2024 Large Language Model Day: Multilinguality in the Era of LLMs.	2024
Google DeepMind: 80/20 Rule in Word Embedding: A Study on Multilingual Vocabulary and Embedding Redundancy in Encoder Models.	2024
WSDM Cup 2023 Overview: Multilingual Information Retrieval Across a Continuum of Languages	2023
Math and Computing Research Discovery Day, University of Waterloo: No Language Left Behind from the Lens of IR	2023

SERVICE

Competition:
FIRE 2023: Cross-Lingual Information Retrieval for African Languages
WSDM Cup 2023: Multilingual Information Retrieval Across a Continuum of Languages

Area Chair:
ARR 2023

Conference Reviewer:
ARR 2023-Present; SIGIR 2024; ECIR 2023

Conference Volunteer:
EMNLP 2022; ACL 2023; NAACL 2024

Departmental Service:
Data System Group (DSG) Lab Committee Member, University of Waterloo (2022–)

TEACHING EXPERIENCE	TA, CS 135 Designing Functional Programs	Fall 2022
	TA, CS 136 Elementary Algorithm Design and Data Abstraction	Winter 2021
		Spring 2022, Winter 2023
	TA, CS 246 Object-Oriented Software Development	Spring 2021
	TA, CS 343 Concurrent and Parallel Programming	Fall 2021, Winter 2022
	TA, CS 479/679 Neural Network	Winter 2025
AWARDS	Outstanding Paper Award at EMNLP 2024	2024
	Best Paper Nominee at SIGIR 2024	2024
	David R. Cheriton Graduate Scholarships in Computer Science Award	2023
	International Award of Excellence Scholarship	2020